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Impact of electrolyte impurities and SEI composition on battery safety†

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Li-ion batteries have a potential risk of thermal runaway. Current safety evaluations in academia and industry rely on experiments or semi-empirical simulations. This limits the understanding of processes leading to or occurring during thermal runaway and how chemical species and impurities can impact them. The limited (quantitative) understanding in turn hinders a holistic safety assessment and optimisation of countermeasures through design or operation. The here presented thermal-runaway model contains a detailed degradation reaction network, which allows the impact of chemical species and impurities on thermal runaway to be studied. We set a particular focus on water impurities and solid-electrolyte interphase (SEI) properties, as both are known to impact life-time of batteries. SEI composition and thickness change during ageing, which is shown here to impact battery safety significantly. The model can reproduce reported experimental behaviour: aged cells are more safe, as they start self-heating, *i.e.* heat production without an external heat source, at 15–20 °C higher temperatures than fresh cells. Our model suggests a thick inorganic and thus less reactive SEI as the underlying cause. Furthermore, we could show that extensive electrode drying to remove water impurities before building battery cells will not significantly improve safety characteristics. In contrast, electrodes not subjected to any drying procedure cause an earlier start of the self-heating phase, *i.e.* have a higher risk of thermal runaway. These insights into the sensitivity to thermal runaway allow robust methods to be tailored for its prevention, from controlling battery and SEI properties during production to adjusting safety assessment for effects of ageing.

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1 Introduction

The increase in electromobility demands cost-effective and safe Li-ion battery materials.[1] With ever-increasing energy densities, safety aspects in Li-ion batteries receive more and more attention from researchers.[2] The most critical safety issue with Li-ion batteries is thermal runaway. This phenomenon of uncontrolled temperature increase in the battery starts with slow self-heating, *i.e.* heating without external heat sources, and can end in explosions, fires, and cell temperatures above 800 °C.[3]

Especially the initial period of self-heating provides an opportunity to apply counteracting measures to prevent thermal runaway. Yet, designing effective countermeasures demands a thorough understanding of the processes triggering and occurring during this critical initial period. In a previous study, we could show that the self-heating phase is characterised by an intricate balance between endothermic and exothermic processes.[4] The main exothermic reactions are related to the decomposition and reformation of the solid electrolyte interphase (SEI), a protective layer at the surface of carbon-based negative active material particles.[4,5] Presently, not much is known about the effect of SEI properties and highly reactive impurities such as water on the self-heating behaviour of Li-ion batteries. This motivates us to conduct a detailed analysis in this publication.

Since its discovery in the late 1980s, the SEI is among the most investigated parts of Li-ion batteries.[6–8] As a naturally formed surface layer on the carbon electrode, it originates from solvent and conductive salt reduction during the first charge–discharge cycle, *i.e.*, during the so-called cell formation process. After formation, the SEI serves as an electric isolator in the ion-conducting phase. Stabilising the SEI and preventing its continuous growth due to ongoing SEI reactions is crucial for reaching long battery life. Tremendous efforts have been undertaken to learn more about its composition, structure, and effects on performance.[7–9] In a recent study, Liu *et al.*[10] combined electrochemical measurements, an electrochemical quartz crystal microbalance, atomic force microscopy, X-ray photoelectron spectroscopy, and online electrochemical mass spectrometry to estimate the composition and thickness of the SEI after formation. With this combined approach, they were the first to show that the initially formed organic SEI component lithium ethylene dicarbonate (LEDC) is partially oxidised

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to form Li_2O and several gases. However, the influence on battery performance was not investigated in their study. Son *et al.*[11] compared the influence of the SEI-forming additives fluoroethylene carbonate (FEC) and vinylene carbonate (VC) on battery performance. They found that FEC could enable better fast-charging capabilities and long-term capacity retention. This is assumed to be connected to a higher resistance for the cell with VC, which is caused by a more polymeric SEI stemming from VC as an additive. The higher resistances on the electrode surface lead to higher overpotentials at elevated currents, eventually resulting in lithium plating during charging. As the SEI is a nanometer-thin surface layer, experimental investigations are very complex,[12] and quantitative analysis has not yet been possible. Thus, a plethora of simulation methods have been used to tackle this problem.[13]

In a simulation study using a kinetic Monte-Carlo (kMC) approach connected to a Newman-type battery cell model, Röder *et al.*[14] showed that higher currents during initial SEI formation may lead to a gradient in the SEI thickness along the electrode thickness. With a holistic model-based analysis of experimental electrochemical impedance spectroscopy data and C-rate tests, we could relate the ageing behaviour of Li-ion cells, among other factors, to an evolution of SEI thickness and interfacial properties.[15] The model contained a Newman-type cell model extended with an SEI. In a recent study Esmaeilpour *et al.*[16] used a kMC approach with over 50 000 simulation conditions to find that the most probable way of SEI formation is a sequence of formation, dissolution and finally, aggregation. Furthermore, approaches using kMC methods in combination with computational chemistry support long-time existing hypotheses about the composition and morphology of the SEI.[17–19] These studies illustrate how simulations can be used as powerful tools to unravel, supplement or support existing theories on SEI growth and functionality.

It was already 20 years ago when the first experiments by Dahn *et al.* connected the self-heating behaviour of Li-ion cells to SEI decomposition.[5] In experimental studies of aged cells, an increase in the self-heating temperature of about 15–20 °C had been suggested to stem from a thicker and more inorganic SEI.[20–23] However, neither detailed experimental nor simulation-based studies have so far been performed on the influence of SEI composition or thickness on battery safety, especially during the self-heating phase. Thus, this publication aims to tackle this open challenge.

H_2O is a known impurity during battery manufacturing.[24] Hygroscopy of electrodes is the most often referenced origin of water in cells.[25,26] Several studies showed the detrimental effect of water on battery performance.[26–28] However, also positive impacts of H_2O addition for Li-metal batteries have been reported.[29] LiF is formed from the reaction of H_2O with $LiPF_6$, and it is a stable and highly conducting SEI component leading to better performance. In contrast, worse performance is usually connected to formation of the phosphorous decomposition products PF_5 and POF_3. These reactions are known to occur at room temperature and are inherently linked to battery ageing. Weber *et al.*[30] investigated the ageing of an electrolyte mixture stored at 95 °C and identified 12 different organo-phosphoric decomposition products. They suggested that a detailed analysis of the formation of these products during ageing could be used to identify "ageing stages" for $LiPF_6$-based electrolyte composition. Stich *et al.*[31] investigated the kinetics of these decomposition reactions by purposely contaminating an electrolyte solution with 1000 ppm water and measuring the concentrations of H_2O, HF and HPO_2F_2. They found that hydrolysis is not following a simple rate law and thus developed a kinetic model to describe their experiments. Huttner *et al.*[32] studied the influence of different drying strategies for water removal on battery performance. They found that extensive drying can decrease performance. This is due to the extreme conditions the battery materials are subjected to during drying. In general, H_2O contamination and follow–up reactions might not immediately influence battery performance, as Zheng *et al.* showed.[28] They found that after 100 cycles, the capacity retention for batteries contaminated with H_2O and without contamination was in the same range, with around 95% remaining capacity for water-free and 90% remaining capacity for water-containing batteries. However, after 300 cycles, increased H_2O content drastically reduced capacity retention. The results show 90% remaining capacity for water-free batteries and 55% for water-contaminated ones. The water-containing battery had 14 mg water added to an 18650 battery. Despite these efforts to qualitatively and quantitatively correlate H_2O contamination and battery performance, we have not found any reports in the literature on tests to investigate the effect of water on battery safety, especially during the crucial self-heating phase. This contribution will shed light on the extent to which H_2O contamination influences battery safety.

To address both the sensitivity of water impurities and of SEI composition to thermal self-heating and thermal runaway, we use a component-based Li-ion battery degradation model. This approach allows us to assess the impact of each participating chemical species on the progression towards thermal runaway at open circuit potential. Our comprehensive model encompasses 12 decomposition reactions and 20 participating species, which will be parameterised through two separate experiments. First, for the conductive salt decomposition, the experiments conducted by Stich *et al.*[31] will be used. The complete set of reactions is then parameterised against an accelerated rate calorimetry (ARC) measurement.[33] The thus parameterised model is then used to conduct in-depth case studies on the effect of LEDC content in the SEI, the SEI thickness, and the H_2O contamination on self-heating and thermal runaway. Eventually, we present a broader parameter study that combines all three effects, examining potential interdependencies between them and illustrating dominant processes and properties.

2 Methods and study cases

This section outlines the underlying reaction network. Furthermore, it provides a detailed description of the model with its assumptions, and the experiments employed for model parameterisation and validation. Eventually, the case study as

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well as the initial conditions of the reference case are introduced.

2.1 Reaction network

The reaction network of SEI and electrolyte degradation forms the core of this comprehensive case study as it allows for the traceback of thermal behaviour to single reactions, reaction interactions and species. The subsequent section provides a thorough overview of the postulated reactions, citing the relevant literature and discussing explicit and implicit interactions. Table 1 summarises the reactions, including the abbreviations used in this work, their reaction enthalpies and the prospective temperature range of a notable start of each reaction.

Considering thermal battery decomposition, the first reactions that occur during heating of the cells at notable rates are the conductive salt decomposition (CSD) and the subsequent decomposition to POF_3 (PFD) and HPO_2F_2 (POFD). They are reported to occur at elevated temperatures around 60 °C to 80 °C. They also happen at room temperature, but only slowly and in the range of days, much longer than during thermal abuse, where reactions occur within hours or minutes. Following the publication of Stich *et al.*,[31] we consider the decomposition of $LiPF_6$ to PF_5 and LiF (CSD) and the subsequent decomposition of PF_5 with H_2O to POF_3 and HF (PFD) as equilibrium reactions. In their work,[31] PF_5 directly reacts with two H_2O to form three HF and HPO_2F_2. However, in a study by Solchenbach *et al.*,[34] POF_3, an intermediate, was detected by online electrochemical mass spectrometry. This detection implies that the lifespan of POF_3 is sufficiently long to be considered significant and cannot be disregarded in our analysis. Thus, we consider the first decomposition of PF_5 with H_2O to form POF_3 and HF (PFD) and the subsequent decomposition of POF_3 with H_2O to form HF and HPO_2F_2 (POFD) as individual reactions.

The decomposition of $LiPF_6$ and the subsequent reactions of its decomposition products PF_5 and POF_3 initiate the release of HF, which causes the breakdown of Li_2CO_3 by reacting with HF to form LiF, H_2O, and CO_2 (ISD). Freiberg *et al.*[35] reported that Li_2CO_3 coated on a carbon electrode reacted immediately with protons, and thus HF, to release CO_2. From this, we conclude that the decomposition reaction of Li_2CO_3 with HF or H^+ is not rate-limiting and that it will happen rapidly as soon as $LiPF_6$ is decomposed and releases HF.

The decomposition of the organic SEI component LEDC (OSD) is reported to start somewhere in a wide temperature range from 60 °C to 120 °C.[5] It is responsible for the transition into the self-heating phase of an ARC measurement.[5,36]

The decomposition of LiOH with lithium metal or Li intercalated in graphite to form H_2 and Li_2O (LSD) is frequently reported in the literature.[25] However, we could not find any information on the temperature range for this reaction. Thus, we depend on a study of the decomposition of LiOH with LiH into Li_2O and H_2, which reported a decomposition temperature around 120 °C.[37]

The decomposition of SEI compounds, including Li_2CO_3, LEDC, and LiOH, eventually exposes the bare electrode surface. Given the solvent's instability against the anode's low potentials, subsequent SEI formation reactions occur at the electrode. These SEI-producing reactions (OSP, ISP, and LSP) are direct results of the preceding decomposition and are linked to the battery's self-heating.[31]

All reactions so far occur already below 130 °C and involve species in the anode or electrolyte. At significantly higher temperatures above 150 °C, self-heating is strongly accelerated by the decomposition of the cathode active material $LiCoO_2$ (CD). The released O_2, in turn, leads to the highly exothermic combustion of solvent molecules (ECD and EMCD),[36] further accelerating thermal self-heating.

Table 1 Degradation reactions, their reaction enthalpy (blue = endothermic, red = exothermic) and the temperature range at which they are observed in differential scanning calorimetry, accelerated rate calorimetry or electrochemical measurements. Products that are also reactants are highlighted in bold[a]

Name	Abbr.	Equation	$\Delta_r H$/kJ mol^{-1}	T_{Start}/°C
Conductive salt decomposition[29]	CSD	$LiPF_6 \rightleftharpoons LiF + \mathbf{PF_5}$	82.07	25*/60–80
PF_5 decomposition[29]	PFD	$PF_5 + H_2O \rightleftharpoons 2\mathbf{HF} + \mathbf{POF_3}$	80.09	25*/60–80
POF_3 decomposition[29]	POFD	$POF_3 + H_2O \rightarrow \mathbf{HF} + HPO_2F_2$	296.53	25*/60–80
Organic SEI production[30,31]	OSP	$2LiC_6 + 2C_3H_4O_3\ (EC) \rightarrow \mathbf{(CH_2OCO_2Li)_2} + C_2H_4 + 2C_6$	-135.73	25**/80–120
Inorganic SEI production[30,31]	ISP	$2LiC_6 + C_3H_4O_3\ (EC) \rightarrow \mathbf{Li_2CO_3} + C_2H_4 + 2C_6$	-572.62	25**/80–120
LiOH production[20]	LSP	$LiC_6 + H_2O \rightarrow \mathbf{LiOH} + 0.5H_2 + C_6$	-199.11	25**/80–120
Organic SEI decomposition[5,31]	OSD	$(CH_2OCO_2Li)_2 \rightarrow \mathbf{Li_2CO_3} + C_2H_4 + CO_2 + 0.5\mathbf{O_2}$	-187.05	60–120
Inorganic SEI decomposition[32]	ISD	$Li_2CO_3 + 2HF \rightarrow 2LiF + \mathbf{H_2O} + CO_2$	-152.09	25
LiOH decomposition[33]	LSD	$LiC_6 + LiOH \rightarrow Li_2O + 0.5H_2 + C_6$	-113.82	100–120
Cathode decomposition[34]	CD	$Li_zCoO_2 \rightarrow zLiCoO_2 + \frac{1-z}{3}\mathbf{O_2} + \frac{1-z}{3}Co_3O_4$	-240.10	150–220
EMC combustion[34]	EMCD	$3.5O_2 + C_4H_8O_3\ (EMC) \rightarrow 4CO_2 + 4\mathbf{H_2O}$	-2071.7	180–350
EC combustion[34]	ECD	$2.5O_2 + C_3H_4O_3\ (EC) \rightarrow 3CO_2 + 2\mathbf{H_2O}$	-1161.3	180–350

[a] * slow process during cell formation; ** process considered indirectly.